

ShadowFox — Data Science Internship

Intern: [Your Name]

Organization: ShadowFox

Domain: Data Science

Duration: [Start Date] – [End Date]

Repository Structure

```
ShadowFox/
|
├── Beginner/
|   ├── Visualization_Library_Documentation.pdf ← Task 1 deliverable
|   └── README.md
|
├── Intermediate/
|   ├── Twitter_Sentiment_Analysis.ipynb ← Task 2 Jupyter Notebook
|   ├── twitter_sentiment_analysis.py ← Task 2 Python script
|   └── README.md
|
├── Advanced/
|   ├── IPL_Fielding_Analysis.xlsx ← Task 3 deliverable
|   └── README.md
|
└── README.md ← (this file)
```

Task 1 — Visualization Library Documentation (Beginner)

Objective

Create a comprehensive documentation guide for 2 Python visualization libraries.

Libraries Covered

Library	Version
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		Purpose
Matplotlib	Latest	Low-level, highly customizable 2D plotting
Seaborn	Latest	High-level statistical visualization on top of Matplotlib

What's Documented

- Library overview & key features
- **14 chart types** across both libraries — each with description, use case & code snippet
- Line, Bar, Histogram, Scatter, Pie, Area (*Matplotlib*)
- Scatter, Line, Bar, Count, Box, KDE, Heatmap, Regression (*Seaborn*)
- Full comparison table: ease of use, customisation, interactivity, performance

Deliverable



Beginner/Visualization_Library_Documentation.pdf

✅ Task 2 — Twitter / X Sentiment Analysis (Intermediate)

Objective

Perform sentiment analysis on Twitter/X posts using **VADER** to classify tweets and identify sentiment trends.

Tools & Libraries

`vaderSentiment` `pandas` `matplotlib` `seaborn` `re`

Research Question

What is the overall sentiment distribution of public tweets, and can we identify patterns in how positively or negatively people express themselves on X?

Methodology

1. Load & explore tweet dataset

2. Text preprocessing (URL removal, mention cleaning, hashtag normalisation)
3. VADER scoring → compound score → classify as Positive / Neutral / Negative
4. Visualise with 5 charts (Pie, Bar, Histogram, Box Plot, Stacked Bar, KDE)
5. Statistical summary & conclusions

Key Results

Sentiment	Count	Avg Compound
Positive	15	+0.72
Negative	15	-0.61
Neutral	15	+0.01

Deliverables

-  Intermediate/Twitter_Sentiment_Analysis.ipynb
-  Intermediate/twitter_sentiment_analysis.py

How to Run

```
pip install vaderSentiment pandas matplotlib seaborn
python twitter_sentiment_analysis.py
# or open Twitter_Sentiment_Analysis.ipynb in Jupyter
```

✓ Task 3 — Cricket Fielding Analysis (Advanced)

Objective

Conduct detailed fielding performance analysis for **7 players** from IPL Match IPL2367 (1st Innings — Delhi Capitals).

Performance Score Formula

$$PS = (CP \times 1) + (GT \times 1) + (C \times 3) - (DC \times 3) + (ST \times 3) + (RO \times 3) - (MRO \times 2) + (DH \times 2) + RS$$

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Code	Metric	Weight
CP	Clean Picks	+1
GT	Good Throws	+1
C	Catches	+3
DC	Dropped Catches	-3
ST	Stumpings	+3
RO	Run Outs	+3
MRO	Missed Run Outs	-2
DH	Direct Hits	+2
RS	Runs Saved/Conceded	As-is

Workbook Structure (5 Sheets)

Sheet	Contents
Overview	Match info, formula reference, weight table
Fielding Data	Ball-by-ball records for 12 deliveries
Performance Matrix	Player × metric grid with totals
PS Calculations	Step-by-step PS computation per player
Charts & Analysis	4 charts + 6 key strategic insights

Top Performers

- 🏆 **Yash Dhull & Axer Patel** — PS = 11 pts (joint best)
- 🎯 **Most Runs Saved:** Kuldeep Yadav (+4)
- 👏 **Most Clean Picks:** Aman Khan (4 CP)

Deliverable

 Advanced/IPL_Fielding_Analysis.xlsx

 Pre-requisites Completed

- LinkedIn profile updated with "ShadowFox Data Science Intern"
 - GitHub repository named "ShadowFox" created
 - All tasks completed as per task list
 - LinkedIn video explanation posted
 - Screenshots submitted with task submission
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